

Question	Answer	Marks
3(a)	resultant force (in any direction) is zero	B1
	resultant moment / torque (about any point) is zero	B1
3(b)(i)	$F = \rho Vg$ $V = 93\,000 / (1.2 \times 9.81)$ $= 7900 \text{ m}^3$	A1
3(b)(ii)	weight = $93\,000 + 3(.0) \times 10^3$	C1
	$m = (93\,000 + 3.0 \times 10^3) / 9.81$ $= 9800 \text{ kg}$	A1
3(c)(i)	$(\Delta p) = F(\Delta t)$ or $F = \Delta p / (\Delta t)$ or $F = \text{rate of change of momentum}$ or $F = m(v - u) / t$	C1
	$\Delta p = 2800 \times 0.50$ $= 1400 \text{ kg m s}^{-1}$	A1
3(c)(ii)	$(\Delta p =) mv$	C1
	$v = 1400 / 64$ $= 22 \text{ m s}^{-1}$	A1

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Question	Answer	Marks
3(c)(iii)	$E_k = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 64 \times 22^2$	A1
	$= 15\,000\text{ J}$	
	or	
	$E_k = p^2 / 2m$	(C1)
	$= 1400^2 / (2 \times 64)$	(A1)
	$= 15\,000\text{ J}$	

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Question	Answer	Marks
1(c)(i)	universal constant	D1